Milestone 3 – Final White Paper

Predicting Four-Week Store-Level Liquor Purchases from Transaction History

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Business Problem and Research Questions

Beverage and alcohol distributors need to commit warehouse space and working capital before retail accounts order. When a commitment misses, capital sits in slow stock or a fast-moving item runs short. Most small and medium-sized distributors still plan with spreadsheets, manual counts, and staff memory. This paper tests whether the order history a distributor already holds can forecast purchases at the level where replenishment decisions are made.

Three research questions and hypotheses guide the analysis:

1. **Q1.** Can purchase history predict the bottles a store will buy of a product over the next four weeks more accurately than naive baselines? *(H1: history-based models beat naive carry-forward on MAE.)*
2. **Q2.** Which product, store, and temporal characteristics are most associated with future purchases? *(H2: recency and frequency features outrank static product attributes.)*
3. **Q3.** Is prediction more dependable for a product within a store or a category within a store? *(H3: accuracy improves as the grain coarsens.)*

The data used is a record of retailer purchases from Iowa's state wholesaler. This does not include inventory, stockouts, or consumer sales, so the models only predict ordering behavior. The results are meant to supply an input to a replenishment decision, not the decision itself.

Background

Iowa is an alcohol control state, meaning the Iowa Department of Revenue is the sole wholesaler of spirits to Class E retailers. Class E retailers include grocery, liquor, and convenience stores. The Department charges every licensee the same price regardless of quantity (Iowa Legislative Services Agency, 2023). All retailer purchases of spirits therefore pass through one wholesaler and are published at the invoice-line level.

Product-by-location demand is hard to forecast because most series are small and intermittent (Fildes et al., 2022). Croston's (1972) method and its bias-corrected variant (Syntetos & Boylan, 2005) remain the standard benchmarks for such series, while gradient-boosted trees trained across many series won the M5 retail competition (Makridakis et al., 2022). In beverage alcohol distribution, Arora et al. (2020) cut error by nearly 50% against a naive model for a distributor's top product, and Ford et al. (2020) forecast at the customer-product level; both found that no single model fits every product. The project connects to Pourtion, a platform in development for distributors, but its question is analytical: whether a predictive capability is worth building.

Data Explanation

Source and Scope

The sole source of data is *Iowa Liquor Sales, 2026* (dataset 1263), published under a CC BY license on the Iowa Data Hub (Iowa Department of Revenue, Alcohol & Tax Operations

Division, 2026). Each row is one invoice line: one product bought by one Class E licensee on one date. The extract covers January 1 to August 31, 2026, and arrived as four CSV parts totaling 1,968,354 rows and 23 columns. After cleaning it holds 82,673 invoices from 2,203 stores in 99 counties, covering 4,429 products in 44 categories and 19.5 million net bottles (Table 1; Appendix A1).

Preparation

Each step is reproduced in the notebook and logged in Appendix A2.

Duplicates: the export contained 364,160 exact duplicate rows, almost all in July (42.7% of rows) and August (50.0%). Removing them left 1,604,194 rows, within 14 of the 1,604,208 the Iowa Data Hub publishes, confirming an export artifact rather than real purchasing (Figure 1).

Missing values: 2,365 rows lacking store geography were filled from the same store's records where possible (743 remained UNKNOWN), and 26 rows lacking a category were labeled UNKNOWN.

Returns: 1,247 negative lines were netted into weekly totals, floored at zero.

Aggregation: lines were summed to store × product × week bottles over the 34 complete Monday-start weeks (January 5 to August 30), with zeros for weeks without a purchase.

Eligibility: only 5.2% of possible store-product pairs appear at all, and the median pair buys in 2 of 34 weeks (Figure 2). A series is therefore forecast at an origin only if it bought in at least 3 of the preceding 8 weeks. Milestone 1 proposed a fixed rule of 12 active weeks, but judging activity over the full calendar would use future weeks (look-ahead) and select series because they happened to be busy in the test period (selection bias). Judging eligibility at each origin from past weeks alone avoids both problems and mirrors how a planner decides which accounts to forecast. On average 30,786 store-product series qualify per origin, covering 36.8% of next-four-week bottles; at the store-category grain, 21,821 qualify and cover 87.5%.

Table 1
Brief Data Dictionary of Key Fields

Field	Type	Role in analysis
ordered_on	Date	Assigns each line to a Monday-start week
store_no	Text ID	Store key for series and store-context features
county_name	Text	Basis for county store density (rural–urban proxy)
item_no	Text ID	Product key (store-product grain)
category_name	Text	Category key (store-category grain) and feature
bottle_volume_ml, pack	Integer	Static product attributes
state_bottle_retail	Decimal (USD)	Median shelf price per product (weeks 1–26)
sales_bottles	Integer	Quantity summed to weekly totals; source of the target
target_4w (derived)	Integer	Bottles bought in the four weeks after the forecast origin

Note. Bottles are the only quantity field intact for the whole period; January prices and volumes are rounded to whole numbers. Full dictionary in Appendix A1.

Methods

Prediction Target and Features

The unit of observation is one series at one forecast origin week *t*; the target is total bottles bought in weeks *t*+1 through *t*+4. Four weeks matches a monthly replenishment rhythm and smooths biweekly ordering, while 34 weeks of history rules out longer horizons. Weekly origins from week 8 to 30 yield 708,074 store-product rows.

Twenty-seven features fall into five families (Appendix A3): series recency and frequency, store context, product market activity, static product attributes, and calendar. The feature function receives the weekly data already truncated at week *t*, and a test that scrambled every later week confirmed that no feature, baseline, or eligibility decision changed.

Validation Design

Validation respects time (Hyndman & Athanasopoulos, 2021). Training origins 8–22 (440,649 rows) have targets ending by July 5; origins 23–25 are an embargo; test origins 26–30 (168,578 rows) have targets from July 6 to August 30. Three expanding-window folds inside training provide rolling-origin cross-validation (Bergmeir & Benítez, 2012), which chose the best model; the test set was used once.

Baselines, Models, and Metrics

Three spreadsheet-level baselines were used: last four weeks carried forward (the H1 benchmark), an eight-week moving average, and Croston-SBA. Three models were compared. **Poisson regression** is the simplest count model and tests whether a linear combination of history helps. A random forest captures non-linear interactions (Breiman, 2001). Gradient boosting with a Poisson loss reflects the M5-winning approach. Per-series ARIMA and deep learning were rejected because 34 sparse points per series are too few and neither learns across series; ordinary least squares was replaced by Poisson regression because it can predict negative quantities.

Metrics are MAE and RMSE in bottles; WAPE (total absolute error ÷ total actual); R^2; bias; and relative MAE (method MAE ÷ naive MAE, below 1 beats naive). WAPE and relative MAE are scale-free, so the two grains can be compared. MAPE is excluded because about a fifth of targets are zero (Hyndman & Koehler, 2006). Models were built in scikit-learn (Pedregosa et al., 2011).

Analysis

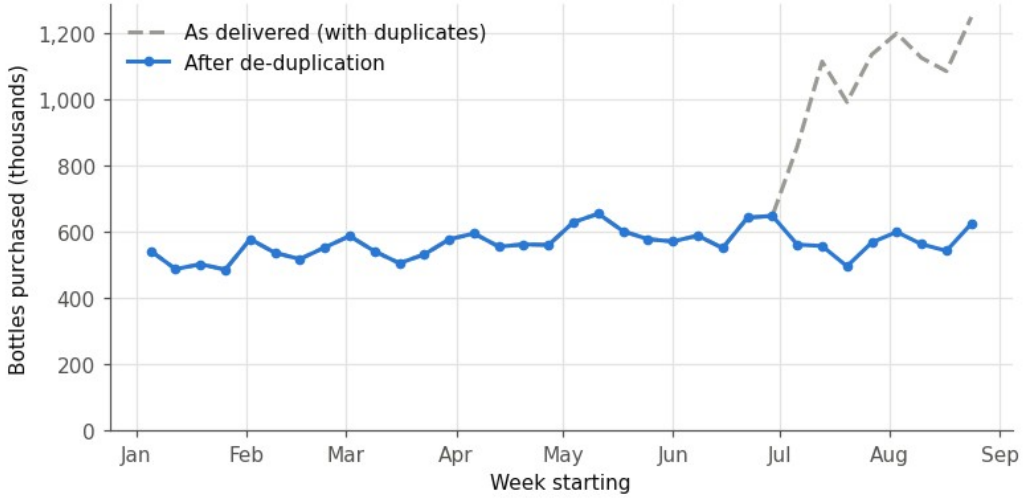
Exploratory Findings

After de-duplication, statewide purchasing is stable at roughly 485,000 to 655,000 bottles per week, with moderate peaks in mid-May and before July 4. The raw export's July–August surge disappears entirely (Figure 1). Demand is concentrated and intermittent with the top 1% of

products accounting for 40% of bottles and the top 10% for 79% (Figure 2a). Most store-product pairs buy only once or twice in 34 weeks (Figure 2b).

Figure 1

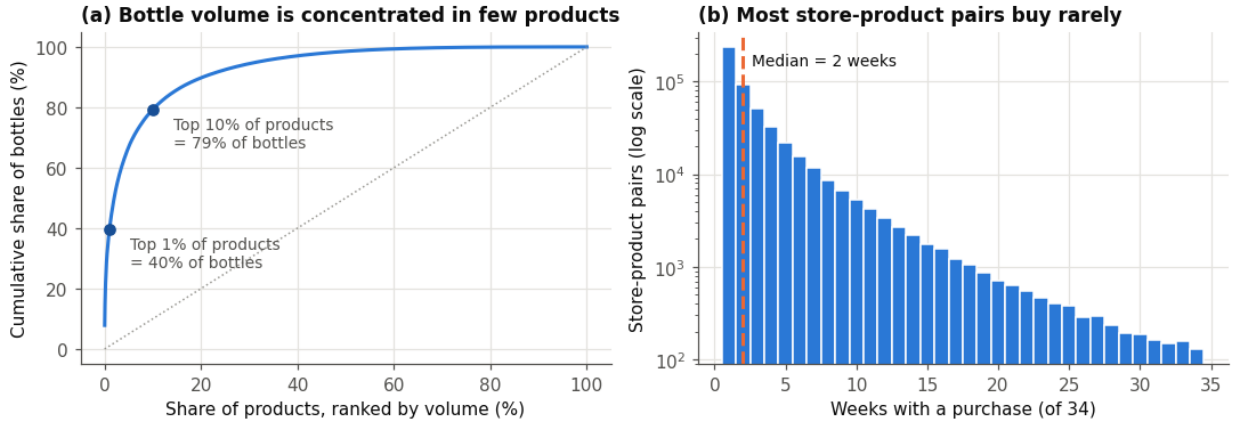
Weekly Bottles Purchased by Iowa Class E Retailers, January–August 2026, Before and After De-duplication



Note. Monday-start weeks, January 5 to August 30, 2026. The dashed line shows the export as delivered; the solid line shows the data after removing 364,160 exact duplicate rows.

Figure 2

Concentration and Intermittency of Demand



Note. (a) Cumulative share of bottles by product rank. (b) Weeks with a purchase, out of 34, for each of the 507,186 store-product pairs that bought at least once (log scale); the dashed line marks the median.

Model Performance and Validation

Table 2 and Figure 3 compare every method on the held-out July–August weeks. At the store-product grain, the cross-validated choice, gradient boosting, reached an MAE of 12.86 bottles and a WAPE of 50.6%, against 16.30 and 64.2% for naive carry-forward: 21% less error (relative MAE 0.79). Every model and smoothed baseline beat the naive rule, so H1 is supported. However, Croston-SBA essentially tied the machine-learning models (MAE 12.84 against 12.75

for the random forest), with the best R^2 (0.62), lowest RMSE, and almost no bias (+1.9%). In cross-validation, gradient boosting had led Croston-SBA (MAE 13.57 ± 0.23 against 14.51 ± 0.58), and its advantage over naive was stable across folds (Appendix A5), but that edge did not survive into the summer test period. Most of the gain comes from smoothing each series' own history, not from machine learning.

Table 2

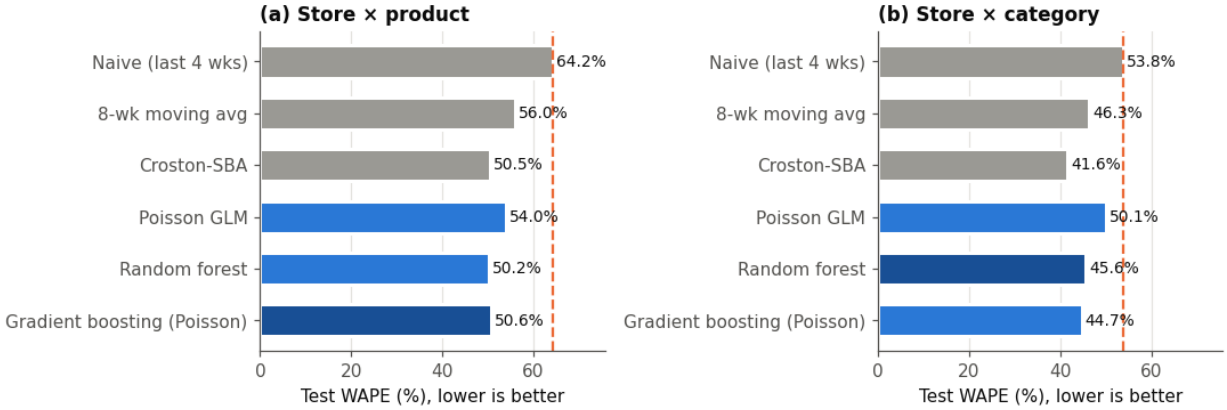
Held-Out Test Performance, Weeks 27–34

Method	SP MAE	SP WAPE	SP Rel. MAE	SC MAE	SC WAPE	SC Rel. MAE
Naive (last 4 weeks)	16.30	64.2%	1.00	46.72	53.8%	1.00
8-week moving average	14.23	56.0%	0.87	40.22	46.3%	0.86
Croston-SBA	12.84	50.5%	0.79	**36.13**	**41.6%**	**0.77**
Poisson GLM	13.73	54.0%	0.84	43.49	50.1%	0.93
Random forest	**12.75**	**50.2%**	**0.78**	39.59†	45.6%	0.85
Gradient boosting (Poisson)	12.86†	50.6%	0.79	38.81	44.7%	0.83

Note. SP = store × product (168,578 test forecasts); SC = store × category (113,028). MAE in bottles per four-week window. WAPE = Σ|error| ÷ Σ actual. Rel. MAE = method MAE ÷ naive MAE. Bold marks the best value in each column; † marks the model selected by cross-validation before testing. RMSE, R^2, and bias appear in Appendix A5.

Figure 3

Forecast Error of Baselines and Models on Held-Out Weeks, by Grain

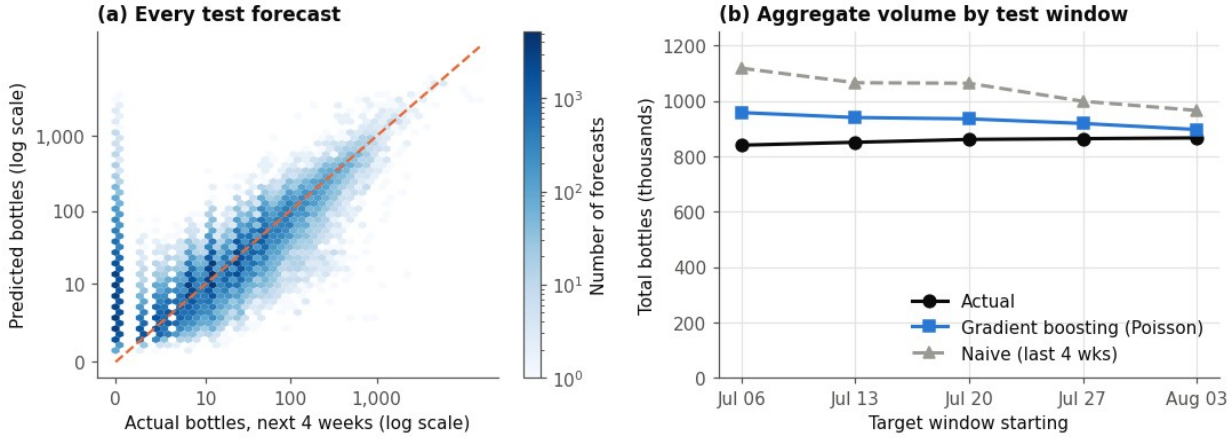


Note. WAPE on test weeks 27–34; lower is better. Gray bars are baselines; the dark blue bar is the model selected by cross-validation; the dashed line marks the naive carry-forward rule.

Forecasts for series above about 20 bottles cluster tightly around the diagonal (Figure 4a); at low volumes the spread widens, and a band at zero reflects the 22% of test windows in which a recent buyer bought nothing. In aggregate (Figure 4b), gradient boosting over-forecast by 8.6% against 21.7% for naive, because series qualify on a burst of recent activity that partly fades in the following weeks. The model beat naive in 59.2% of individual forecasts, and summed to each store's four-week total, WAPE fell to 22.8% against 32.5%.

Figure 4

Actual Versus Predicted Four-Week Purchases, Store-Product Grain, Gradient Boosting



Note. (a) Every test forecast on log scales; the dashed line marks a perfect forecast. (b) Total bottles across eligible series for each test window, with a zero baseline.

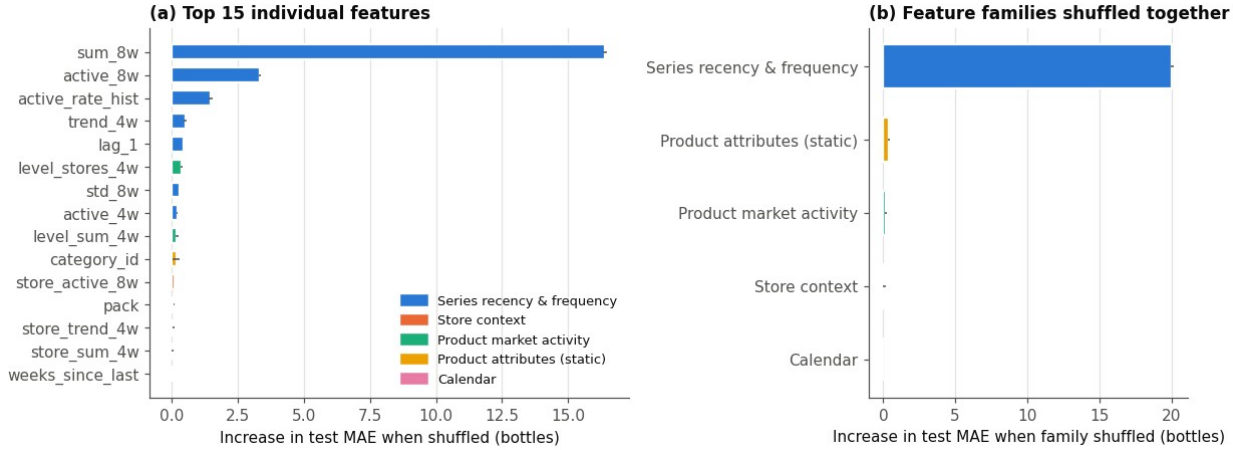
What Drives the Forecasts

Permutation importance measures how much held-out error rises when a feature is shuffled (Breiman, 2001). The eight-week purchase total dominated (MAE +16.4 bottles), followed by active weeks in the last eight (+3.3), the long-run active rate (+1.5), trend (+0.5), and last week's quantity (+0.4; Figure 5a). Shuffling whole families (Figure 5b) raised MAE by 20.1 bottles for series recency and frequency, against 0.39 for static product attributes, 0.20 for product market activity, 0.08 for store context, and zero for calendar features, strongly supporting H2. Correlated features such as the four-week sum show near-zero importance because the eight-week total carries the same information. These are associations, not causes.

Error is not uniform (Appendix A6). The model beat naive in every segment, but WAPE ranged from 78.5% for the lowest-volume quartile to 46.9% for the highest (Figure A1), and was higher for the smallest stores (56.3% against 48–51%) and least dense counties (54.8% against 46.0%).

Figure 5

Permutation Importance of Features and Feature Families, Gradient Boosting



Note. Increase in held-out MAE (bottles) when a feature (a) or a whole family (b) is randomly shuffled; 40,000 test forecasts, five repeats; error bars show one standard deviation.

Aggregation Level

At the store-category grain, absolute accuracy improved: the best WAPE fell from 50.2% to 41.6% and coverage rose from 36.8% to 87.5% of bottles (Table 2, Figure 3b). The margin over naive did not grow (relative MAE 0.77 against 0.78), and Croston-SBA again won, while the cross-validated random forest reached only 0.85. H3 is supported for absolute accuracy and coverage, but not for added value from machine learning.

Conclusion

RQ1: Yes. Purchase history predicts four-week store-product purchases about 21% more accurately than carrying last month forward, but the simple Croston-SBA method matched the machine-learning models out of sample. **RQ2:** A series' own recent volume and purchase frequency drive the forecasts; product attributes, store context, and calendar add little. **RQ3:** Store-product forecasts carry about 50% weighted error, which is useful for review but not dependable on its own; store-category forecasts (42%) and store-level totals (23%) are dependable enough for planning. For a distributor, smoothed, intermittency-aware forecasts reviewed at the account and category level capture most of the achievable gain.

Assumptions, Limitations, and Challenges

Assumptions

Duplicate rows are export artifacts; purchases from the state wholesaler proxy distributor orders; product attributes are stable catalogue information; returns reduce purchases in the week they occur; federal holidays are the only calendar effects known in advance.

Limitations

Eight months contain no full seasonal cycle and the test covers only July–August, so calendar effects and holiday accuracy are unknown. Purchases are not inventory or consumer sales, so nothing here predicts stockouts. Results come from a uniform-price control state and may not transfer to open markets. Eligible store-product series cover only 36.8% of bottles, and the eligibility rule biases most methods toward over-forecasting. Models were lightly tuned, evaluated on one chronological split, and produce no prediction intervals.

Challenges

The duplicate artifact nearly doubled July–August volume and was caught only by reconciling with the portal's row count. The proposal's fixed eligibility rule would have introduced look-ahead and selection bias and was redesigned. The Poisson regression failed to converge with its default solver and was switched to Newton–Cholesky. Finally, the test favored a baseline over the cross-validated choice; this is reported as observed rather than re-selected, which would leak test information into model selection.

Future Uses and Recommendations

Future Uses

With more months, the pipeline could add year-over-year features and a holiday-season test. Joined with a distributor's inventory, lead times, and promotions, forecasts could drive reorder points and safety stock; prediction intervals, a separate model for sporadic pairs, and replication in other markets are natural extensions.

Recommendations

The evidence supports four proportionate steps. First, replace carry-forward planning with a smoothed, intermittency-aware baseline such as Croston-SBA, which is transparent and cut error by about 21%. Second, plan volume at the account and category level (23–42% error) and treat item-level forecasts as prompts for planner review. Third, give extra attention to low-volume items, small stores, and rural counties, where error is highest. Fourth, do not yet deploy the machine-learning models operationally, because their advantage over the best baseline was not confirmed out of sample.

Implementation Plan

The pipeline would run weekly like transaction data → features → forecasts → analytics interface → planner decision. In Pourtion, the distributor platform I am developing, *Phase 1* deploys Croston-SBA forecasts rolled up to account and category totals, flagging large departures from recent ordering. *Phase 2* runs gradient boosting in shadow mode and promotes it only if it beats the baseline by a pre-set margin over at least three months, including the holiday season. *Phase 3* adds inventory and lead times for reorder suggestions. A person places every order.

Ethical Assessment

The data contains no personal information about individuals. Table 3 identifies the relevant concerns, assesses their likelihood and impact in light of the results, and states each mitigation.

Table 3
Ethical Risk Assessment

Identify	Assess (likelihood / impact)	Mitigate
Business re-identification: every store is named and addressed	High / Moderate	No store or business names in figures or tables; examples use anonymized codes (A7)
Geographic and size bias: error is higher for small stores (WAPE 56%) and rural counties (55%) than for large stores and dense counties (46–51%)	Observed / Moderate	Error reported by segment (A6); extra planner review for high-error segments
Data-quality distortion: duplicates, rounded January values, returns	Certain / High	Duplicates removed and reconciled to the portal; returns netted; every rule logged (A2)
Filtering effects: eligible series cover 36.8% of store-product bottles	Certain / Moderate	Coverage stated; conclusions limited to regular buyers
Overreliance: forecasts treated as inventory guidance; the selected model over-forecasts by 8.6%	Moderate / High	Scope boundary stated throughout; bias reported; forecasts framed as advisory
Misleading presentation: aggregates or log scales hide error	Moderate / Moderate	Per-forecast and aggregate views shown together; zero baselines; baselines always shown
Misuse of commercial data if applied to a distributor's accounts	Moderate / Moderate	Use only with data-sharing agreements; aggregate reporting; human review of decisions

Data changes

Every transformation including de-duplication, imputation, netting returns, complete-week restriction, and eligibility is disclosed, reproduced somewhere in the code, and its excluded volume quantified.

Sourcing

The data comes from the official state portal under a CC BY license, attributed in every deliverable, and was verified against the published schema and row count.

Regulation

Alcohol distribution is regulated at the state and federal levels. For full transparency, this project does not attempt to assess compliance, and commercial use of account-level forecasts would need review against applicable trade-practice rules.

Code and Reproducibility

All preparation, modeling, validation, and figures are in the Jupyter notebook *Ludwig_Keaton_Milestone2.ipynb*, submitted with this final paper along with its PDF export. It

uses a fixed random seed (27), includes the leakage tests, ran end to end, and is the direct source of every result reported here.

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Appendix: Supporting Documentation

A1. Extended Data Dictionary

Field	Type	Description
invoice_id	Text	Invoice identifier (82,673 invoices)
ordered_on	Date	Order date (192 dates, January 1–August 31, 2026)
store_no, store_name	Text	Licensee number (2,203 stores) and business name (name never reported)
store_address, store_city, store_zip_code	Text	Store location (473 cities, 508 ZIP codes)
county_fips_code, county_name	Text	County of the store (99 counties)
category_code, category_name	Text	Product category (44 categories)
vendor_number, vendor_name	Text	Supplier of the product (197 vendors)
item_no, im_desc	Text	Product number (4,429) and description
pack	Integer	Bottles per case
bottle_volume_ml	Integer	Bottle size in milliliters
state_bottle_cost	Decimal	State's cost per bottle (USD)
state_bottle_retail	Decimal	Price per bottle to the retailer (USD)
sales_bottles	Integer	Bottles on the line (negative = return/adjustment)
sales_dollars	Decimal	Line value (USD)
sales_liters, sales_gallons	Decimal	Line volume

Note. Counts are from the raw file. Before de-duplication, store geography was missing on 4,129 rows (0.21%) and category on 32 rows.

A2. Data-Cleaning Log

Step	Count
Rows as delivered (four CSV parts)	1,968,354
Exact duplicate rows removed (July 162,590; August 201,557; other months 13)	364,160
Rows after de-duplication (portal publishes 1,604,208)	1,604,194
Rows missing store geography → filled from same store / left UNKNOWN	2,365 → 1,622 / 743
Rows missing category → filled from same item / left UNKNOWN	26 → 0 / 26
Negative-bottle lines netted into weekly totals (bottles)	1,247 (−13,234)
Zero-cost lines (kept)	42
Repeated invoice-item lines (kept)	1,931
Rows outside the 34 complete weeks (excluded from modeling)	23,445
Store-product pairs with any purchase / mean eligible per origin	507,186 / 30,786
Store-category pairs with any purchase / mean eligible per origin	57,499 / 21,821
Share of four-week bottles covered by eligible series (SP / SC)	36.8% / 87.5%

A3. Feature Definitions

Family	Feature(s)	Definition (all computed from weeks ≤ t)
Recency & frequency	lag_1 … lag_4	Bottles in each of the last four weeks
	sum_4w, sum_8w, mean_8w, std_8w	Totals, mean, and standard deviation over the last 4/8 weeks
	active_4w, active_8w	Weeks with a purchase in the last 4/8 weeks
	active_rate_hist	Share of all weeks 1…t with a purchase
	weeks_since_last	Weeks since the most recent purchase
	mean_when_active_8w	Average bottles per purchasing week, last 8 weeks
	trend_4w	log(1 + last 4 weeks) − log(1 + prior 4 weeks)
Store context	store_sum_4w, store_sum_8w, store_active_8w, store_trend_4w	Same measures for the store's purchases across all products
	county_store_count	Number of licensed stores in the store's county
Product market activity	level_sum_4w, level_stores_4w	Statewide bottles and average ordering stores per week for the product (category)
Product attributes	bottle_volume_ml, pack, retail_price, category_id	Median size, pack, and shelf price in weeks 1–26; category
Calendar	target_month, holidays_in_target	Month of the target window's first week; federal holidays inside the window

A4. Modeling Configuration

Item	Setting
Target	Bottles in weeks t+1 … t+4 (four-week horizon)
Origins	Training 8–22; embargo 23–25; test 26–30
Eligibility	At each origin t: purchases in ≥ 3 of weeks t−7 … t
Cross-validation	Expanding window: train ≤ 10 / validate 14–16; ≤ 13 / 17–19; ≤ 16 / 20–22
Croston-SBA	Smoothing α = 0.1; per-week forecast × 4
Poisson GLM	log(1 + x) and standardized numerics, one-hot category; α = 1e-4; Newton–Cholesky solver
Random forest	200 trees, min leaf 20, max features 0.5, 25% bootstrap sample
Gradient boosting	Poisson loss, learning rate 0.05, 400 iterations, 31 leaves, min leaf 50, L2 = 1.0, native categorical
Random seed	27

A5. Additional Metrics and Cross-Validation Results

Table A1

Rolling-Origin Cross-Validation, Mean (SD) Across Three Folds

Method	SP MAE	SP WAPE	SP Rel. MAE	SC MAE	SC WAPE	SC Rel. MAE
Naive (last 4 weeks)	17.19 (0.12)	60.0%	1.00	49.45 (0.28)	51.5%	1.00
8-week moving average	14.88 (0.13)	51.9%	0.87	42.57 (0.47)	44.4%	0.86
Croston-SBA	14.51 (0.58)	50.6%	0.84	42.25 (1.80)	44.0%	0.85

Method	SP MAE	SP WAPE	SP Rel. MAE	SC MAE	SC WAPE	SC Rel. MAE
Poisson GLM	13.83 (0.31)	48.3%	0.81	42.11 (0.66)	43.9%	0.85
Random forest	13.77 (0.24)	48.0%	0.80	**41.68 (0.41)**	**43.4%**	**0.84**
Gradient boosting (Poisson)	**13.57 (0.23)**	**47.4%**	**0.79**	42.11 (0.42)	43.9%	0.85

Note. Bold marks the cross-validated choice at each grain.

Table A2

Held-Out Test RMSE, R², and Bias

Method	SP RMSE	SP R^2	SP Bias	SC RMSE	SC R^2	SC Bias
Naive (last 4 weeks)	80.89	0.389	+21.7%	142.91	0.645	+8.5%
8-week moving average	71.77	0.519	+19.8%	122.44	0.739	+8.7%
Croston-SBA	63.95	0.618	+1.9%	112.62	0.779	+0.4%
Poisson GLM	71.32	0.525	+17.1%	120.74	0.746	+20.3%
Random forest	70.41	0.537	+6.7%	126.86	0.720	+10.5%
Gradient boosting (Poisson)	66.98	0.581	+8.6%	121.06	0.745	+8.2%

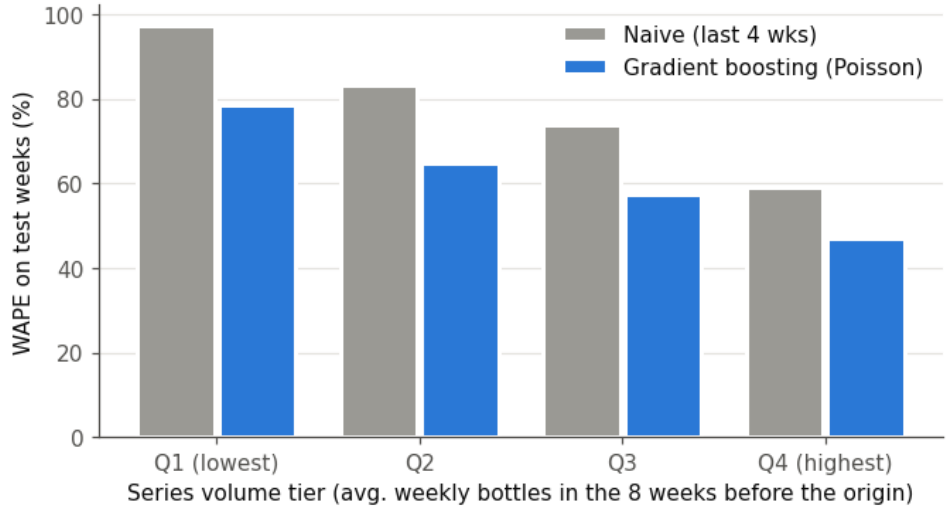
A6. Error by Segment (Store-Product, Gradient Boosting)

Segment	Tier	Forecasts	Mean actual	MAE model / naive	WAPE model / naive
Series volume	Q1 (lowest)	42,145	2.54	1.99 / 2.47	78.5% / 97.4%
	Q2	42,144	7.53	4.88 / 6.26	64.9% / 83.2%
	Q3	42,144	15.41	8.86 / 11.40	57.5% / 74.0%
	Q4 (highest)	42,145	76.14	35.72 / 45.08	46.9% / 59.2%
Store size	Q1 (smallest)	42,158	16.17	9.11 / 12.19	56.3% / 75.4%
	Q2	42,277	21.45	10.32 / 13.39	48.1% / 62.5%
	Q3	42,093	22.81	11.11 / 13.31	48.7% / 58.3%
	Q4 (largest)	42,050	41.24	20.94 / 26.34	50.8% / 63.9%
County density	Low	56,193	18.76	10.28 / 13.22	54.8% / 70.5%
	Medium	56,192	24.23	13.01 / 17.20	53.7% / 71.0%
	High	56,193	33.22	15.29 / 18.49	46.0% / 55.7%

Note. Series volume = average weekly bottles in the eight weeks before the origin; store size = store volume in weeks 1–26; county density = tertiles of licensed stores per county.

Figure A1

Forecast Error by Series Volume Tier, Store-Product Grain



Note. WAPE on test weeks for the naive rule and gradient boosting, by quartile of average weekly bottles in the eight weeks before the origin.

A7. Example Forecasts (Anonymized)

Table A3

Eight Highest-Volume Store-Product Series, Final Test Window (August 3–30, 2026)

Store	Product	Actual	Naive	Gradient boosting
Store 1	Product 1	5,664	4,464	4,569
Store 2	Product 2	4,044	5,340	5,004
Store 3	Product 3	3,960	1,080	1,385
Store 4	Product 4	3,240	1,080	1,855
Store 4	Product 3	2,880	360	2,180
Store 5	Product 2	2,580	2,520	2,921
Store 6	Product 2	2,040	1,440	1,520
Store 7	Product 5	2,016	1,344	746

Note. Bottles over four weeks. Large bulk orders (e.g., Store 4, Product 3) are poorly anticipated by any history-based method, which illustrates why forecasts should prompt review rather than set order quantities.

Ten Questions an Audience Might Ask

These questions were proposed with the Milestone 2 draft. Full answers are provided in the accompanying Q&A document submitted with this paper (Ludwig_Milestone3_QA).

1. Why use Iowa's public liquor data rather than a distributor's own order history, and how representative is a control state with a single wholesaler?
2. Why a four-week horizon instead of next-week or next-quarter forecasts?
3. How do you know no information from the test weeks leaked into training, including through the eligibility rule?
4. If Croston-SBA matched gradient boosting on the test weeks, why build machine-learning models at all?
5. Why were the cross-validation winners not the best methods on the test set, and should the final choice have changed?
6. What happens to the sporadic store-product pairs excluded by the eligibility rule, and how would a distributor forecast them?
7. Why are forecasts less accurate for small stores and rural counties, and what should a distributor do about it?
8. Why do the models over-forecast in aggregate, and how would you correct that bias in practice?
9. How would results change once a full year, including the holiday season, is available?
10. How would Pourtion (my company) use these forecasts in practice, and what safeguards keep planners from over-trusting them?